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Design & Implementation of an MPPT Solar Charge Controller Using Arduino Nano

Course: EEE-316 Power Electronics Laboratory

Project: Design of MPPT-Based Solar Charge Controller

Department: Electrical & Electronic Engineering

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Chapter 1

Introduction

1.1 Background and Motivation

Solar energy has emerged as one of the most promising renewable energy technologies in the twenty-first century. Unlike fossil fuels, sunlight is freely available, clean, and inexhaustible. However, efficiently converting solar radiation into usable electrical energy requires careful power electronics design, particularly with respect to battery charging.

A solar panel does not deliver a fixed voltage or current – its output changes continuously with irradiance (light intensity), cell temperature, and load conditions. Without a smart charge controller, a large fraction of the available solar power is simply wasted, or worse, the battery may be overcharged or damaged. This motivates the development of an **MPPT Solar Charge Controller**: a device that continuously adjusts the operating point of the solar panel so that the maximum possible power is extracted at every instant.

1.2 Objectives

The objectives of this laboratory experiment are:

1. To understand the concept of Maximum Power Point Tracking (MPPT) in photovoltaic (PV) systems.
2. To design and build an Arduino Nano-based MPPT charge controller suitable for a 50 W solar panel and a 12 V lead-acid battery.
3. To implement the *Perturb & Observe* (P&O) MPPT algorithm in embedded C/C++.
4. To design a DC-DC buck converter stage for voltage step-down conversion.
5. To integrate current, voltage, and power measurement with a 20×4 LCD display.
6. To implement battery charging states: *Bulk*, *Float*, and *Off*.
7. To test and evaluate the performance of the complete system experimentally.

1.3 Scope of the Report

This report covers the theoretical background of solar MPPT, hardware circuit design including schematic and component selection, the complete Arduino firmware with detailed explanation, experimental results with output graphs, and a discussion of performance. The report is organized into seven chapters as shown in the table of contents.

Chapter 2

Theoretical Background

2.1 Photovoltaic Cell Characteristics

A photovoltaic (PV) cell operates on the principle of the photovoltaic effect: when photons from sunlight strike the semiconductor junction, electron-hole pairs are generated, creating an electromotive force. The electrical behavior of an ideal PV cell is described by:

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{nV_T} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \tag{2.1}$$

where I_{ph} is the photogenerated current, I_0 is the dark saturation current, n is the ideality factor, $V_T = kT/q$ is the thermal voltage, R_s is the series resistance, and R_{sh} is the shunt resistance.

The resulting I-V and P-V curves are non-linear. The **Maximum Power Point (MPP)** is the unique operating point (V_{mp}, I_{mp}) where the product $P = VI$ is maximum. This point shifts with irradiance and temperature, necessitating a tracking algorithm.

2.2 The MPPT Concept

Consider a 100 W solar panel that nominally produces $V_{oc} = 21\,\mathrm{V}$ and $I_{sc} = 6.2\,\mathrm{A}$. When connected directly to a 12 V battery:

$$P_{direct} = 12\,\mathrm{V} \times \frac{12}{21} \times 6.2\,\mathrm{A} \approx 42.5\,\mathrm{W} \tag{2.2}$$

This means over 57% of the available power is lost. An MPPT controller uses a DC-DC converter to present the panel with its optimal impedance so that $V_{panel} = V_{mp}$, while delivering the appropriate voltage to the battery. The efficiency gain can be 30–40% in practical systems.

2.3 Perturb and Observe Algorithm

The **Perturb & Observe (P&O)** method is the most widely used MPPT algorithm due to its simplicity and low cost of implementation. The algorithm works as follows:

1. Measure panel voltage V and current I; compute power $P = VI$.
2. Perturb the operating point by a small step ΔD in the PWM duty cycle.
3. Measure the new power P'.
4. If $P' > P$: continue perturbing in the same direction.
5. If $P' < P$: reverse the perturbation direction.

The algorithm oscillates around the MPP with a step-size-dependent steady-state error. In this project the PWM increment `PWM_INC = 1` (1%) provides a balance between tracking speed and power ripple.

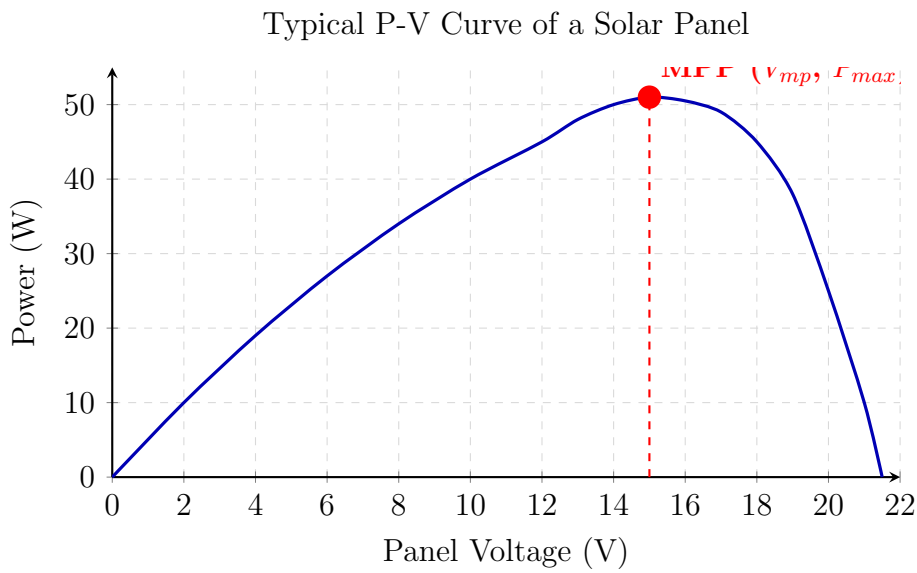

Figure 2.1: Typical P-V characteristic curve of a solar PV panel showing the Maximum Power Point

2.4 Buck Converter Operation

The DC-DC conversion stage is a synchronous buck converter. Its steady-state voltage conversion ratio is:

$$\frac{V_{bat}}{V_{sol}} = D \tag{2.3}$$

where D is the duty cycle ($0 \leq D \leq 1$). The Arduino-generated 50 kHz PWM signal ($T_{pwm} = 20\,\mu$s) drives the MOSFET gate through an IR2104 half-bridge gate driver. The inductor current ripple is:

$$\Delta I_L = \frac{(V_{sol} - V_{bat}) \cdot D}{f_{sw} \cdot L} \tag{2.4}$$

For $L = 33\,\mu$H, $f_{sw} = 50\,$kHz, $V_{sol} = 19\,$V, $V_{bat} = 12\,$V, and $D \approx 0.63$:

$$\Delta I_L = \frac{(19 - 12) \times 0.63}{50 \times 10^3 \times 33 \times 10^{-6}} \approx 2.67\,\text{A} \tag{2.5}$$

2.5 Battery Charging Profile

Lead-acid batteries require a multi-stage charging strategy to maximize capacity and lifespan:

Table 2.1: Lead-Acid Battery Charging Stages

Stage	Voltage Range	Description
Off	$< 11.0\,$V or no solar	MOSFETs off; no charging
Bulk	$11.0 - 13.6\,$V	Maximum current; MPPT algorithm active
Float	$13.6\,$V	Voltage clamped; current tapers to maintain full charge
LVD	$< 11.5\,$V	Load disconnected to prevent deep discharge

Chapter 3

Circuit Design and Hardware

3.1 System Block Diagram

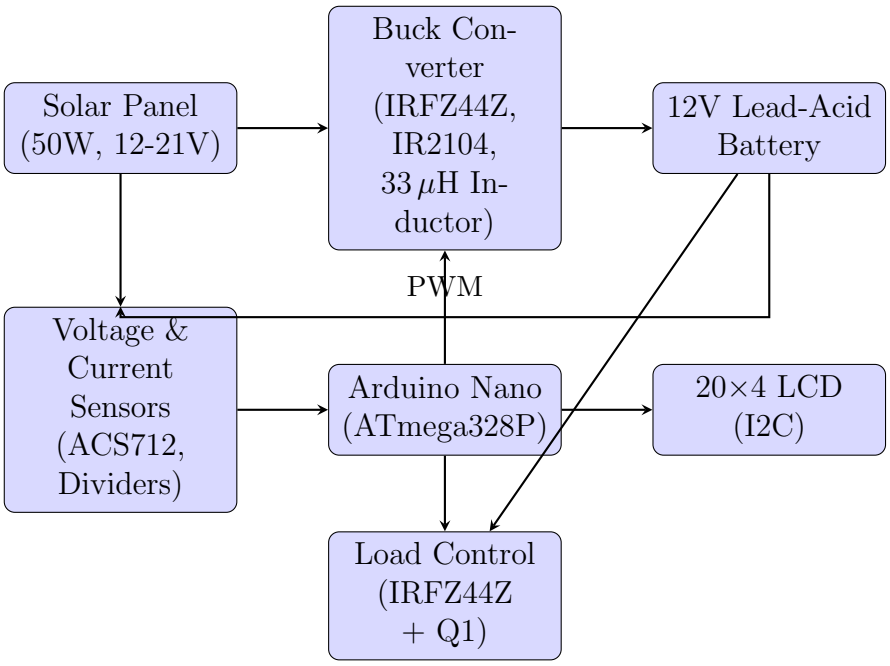

Figure 3.1: System-level block diagram of the MPPT Solar Charge Controller

3.2 Circuit Schematic

The complete schematic is shown in Figure 3.2. The key sub-circuits are described in the following subsections.

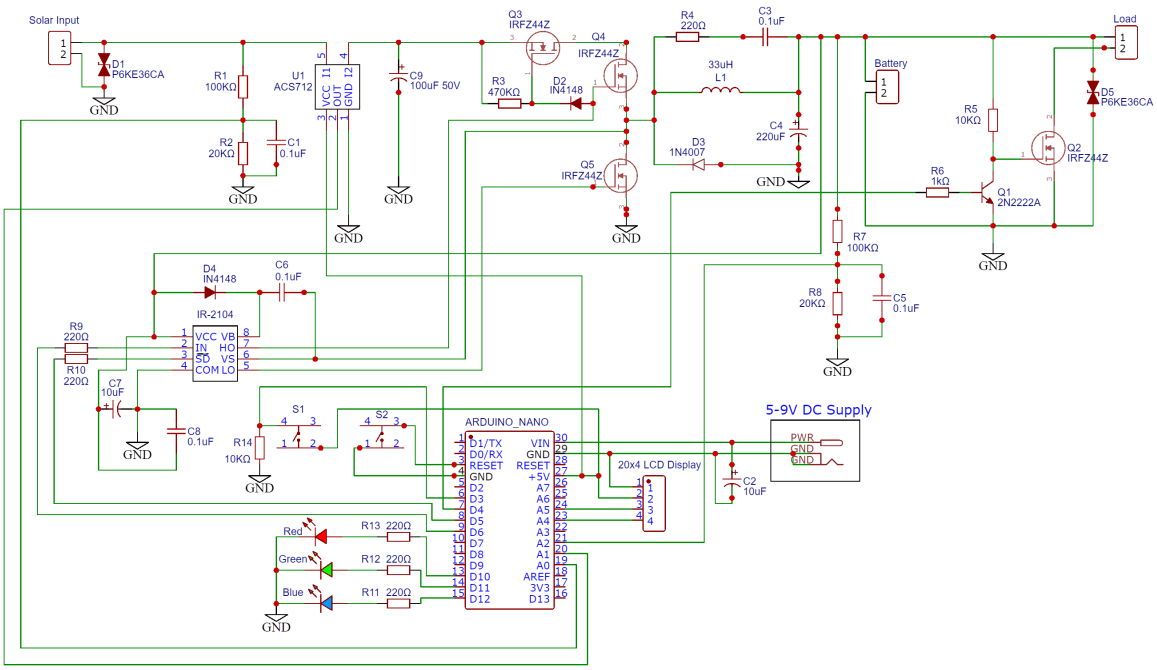

Figure 3.2: Complete schematic of the Arduino MPPT Solar Charge Controller

3.3 Component Description and Selection

3.3.1 Microcontroller – Arduino Nano (ATmega328P)

The Arduino Nano serves as the central processing unit. It features:

- 16 MHz clock, 32 KB Flash, 2 KB SRAM
- 6 analog input pins (10-bit ADC), 6 PWM outputs
- I2C interface for the LCD module
- UART for serial data logging

The TimerOne library is used to generate a 50 kHz PWM signal on pin D6.

3.3.2 Power MOSFETs – IRFZ44Z

Three IRFZ44Z N-channel MOSFETs (Q3, Q4, Q5) form the buck converter switch. Key specs:

- $V_{DS(max)} = 60\,\text{V}$; $I_D = 50\,\text{A}$ continuous
- $R_{DS(on)} = 14.5\,\text{m}\Omega$ (fully enhanced)
- Aluminium heat sinks are attached for thermal management

3.3.3 Gate Driver – IR2104

The IR2104 half-bridge driver (U2) provides bootstrapped high-side and low-side gate drive, preventing shoot-through. Pin SD (enable) is driven by Arduino pin D5 (`PWM_ENABLE_PIN`).

3.3.4 Current Sensor – ACS712

The ACS712-05B Hall-effect current sensor measures solar panel current (0–5 A range). It outputs 185 mV/A centered at 2.5 V. The Arduino reads the analog output on pin A1, mapped using `mapf()` to convert raw ADC to amperes.

3.3.5 Voltage Sensing

Two resistive voltage dividers (R1+R2 and R7+R8) scale the solar panel voltage (∼0–21 V) and battery voltage (∼0–15 V) down to 0–5 V for the Arduino ADC:

$$V_{adc} = V_{input} \times \frac{R_2}{R_1 + R_2} = V_{input} \times \frac{20\text{k}}{100\text{k} + 20\text{k}} = \frac{V_{input}}{6} \tag{3.1}$$

The scaling constant in firmware is:

$$\texttt{SOL_VOLTS_SCALE} = \frac{5}{1023} \times 6 \approx 0.02930\,\text{V/LSB} \tag{3.2}$$

3.3.6 Inductor and Capacitors

- L1: 33 μH toroidal power inductor (visible in the PCB photograph)
- C9: 100 μF/50 V electrolytic input filter capacitor
- C4: 220 μF electrolytic output filter capacitor
- C3: 0.1 μF ceramic snubber capacitor

3.3.7 Protection Devices

- D1, D5: P6KE36CA TVS diodes – transient voltage suppression on solar and load inputs
- D3: 1N4007 freewheeling diode for the buck converter
- D2: 1N4148 fast-recovery diode in the gate drive bootstrap circuit

3.3.8 Load Control Circuit

Q2 (IRFZ44Z) switches the DC load via a 2N2222A NPN transistor (Q1) controlled by Arduino digital pin D4 (`LOAD_PIN`). This provides isolation between the 5 V logic and the battery-voltage load rail.

3.3.9 LED Indicators

Three LEDs provide a visual battery status indicator:

- **Red** (D10): Battery < 11.9 V (low charge)
- **Green** (D11): Battery $11.9 - 14.1$ V (normal)
- **Yellow** (D12): Battery > 14.1 V (fully charged)

3.4 PCB and Prototype Photograph

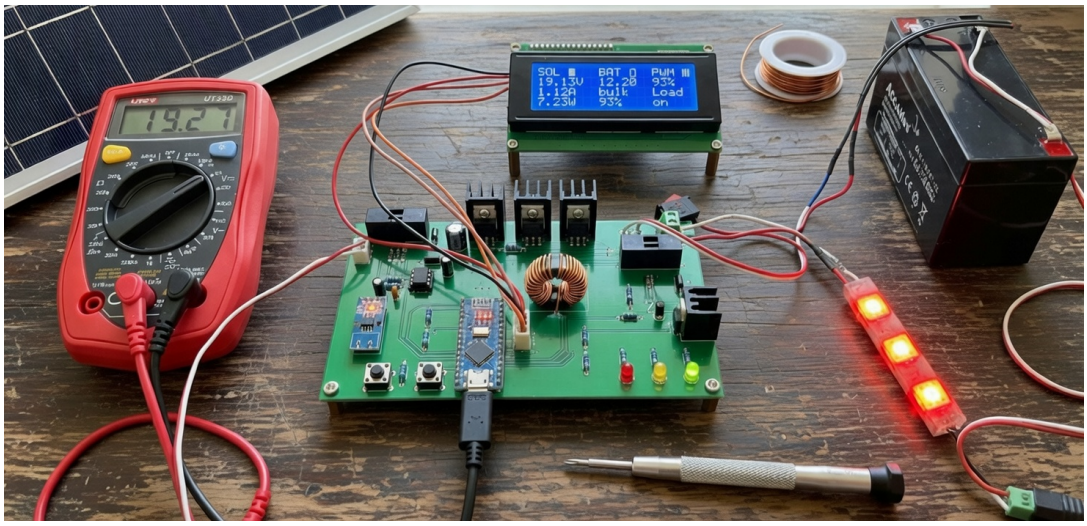

Figure 3.3: Prototype of the MPPT Solar Charge Controller on custom PCB during testing. The multimeter reads 19.27 V solar panel voltage; the LCD shows 19.13 V (panel), 12.20 V (battery), 1.12 A, 7.23 W, 93% PWM, Bulk charging mode.

Chapter 4

Firmware Description

4.1 Software Architecture

The firmware follows a cooperative multitasking pattern. The main loop executes six functions in sequence every iteration:

1. `read_data()` – ADC sampling (solar volts, solar amps, battery volts)
2. `run_charger()` – state machine update (off/on/bulk/float)
3. `print_data()` – serial debug output
4. `load_control()` – load switching logic
5. `led_output()` – LED status update
6. `lcd_display()` – LCD refresh

A hardware interrupt driven by Timer1 counts 50 kHz ticks and increments the seconds counter every 50,000 ticks (i.e., every real second).

4.2 ADC Averaging

Eight ADC samples are averaged per measurement cycle to reduce noise:

```
int read_adc(int channel)
{
  int sum = 0;
  for (int i = 0; i < AVG_NUM; i++)    // AVG_NUM = 8
  {
    sum += analogRead(channel);
    delayMicroseconds(50);
  }
  return (sum / AVG_NUM);
}
```

Listing 4.1: ADC averaging function

4.3 Charger State Machine

The charging algorithm is implemented as a four-state finite state machine with transitions driven by measured voltage and power thresholds.

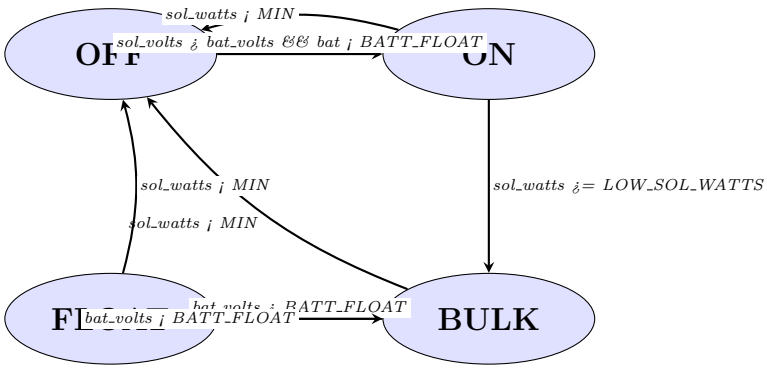

Figure 4.1: Charger state machine transitions

4.4 Complete Arduino Source Code

The full firmware is listed below. Comments have been added to assist understanding.

```
#include "TimerOne.h"
#include <LiquidCrystal_I2C.h>
#include <Wire.h>
#define ENABLE_DATALOGGER 0
#define LOAD_ALGORITHM 1

//        ADC Channel Assignments

#define SOL_AMPS_CHAN  1          // ACS712 current sensor on A1
#define SOL_VOLTS_CHAN 0          // Solar voltage divider on A0
#define BAT_VOLTS_CHAN 2          // Battery voltage divider on A2
#define AVG_NUM        8          // Number of ADC samples to average

//        Voltage Scaling Factors

#define SOL_VOLTS_SCALE 0.029296875  // (5V/1023) * 6 (divider ratio)
#define BAT_VOLTS_SCALE 0.029296875

//        PWM Configuration

#define PWM_PIN        6          // Timer1 PWM output
#define PWM_ENABLE_PIN 5          // IR2104 SD pin (HIGH = enable)
#define PWM_FULL       1023       // 100% duty cycle raw value
#define PWM_MAX        100        // Maximum PWM percentage
#define PWM_MIN        60         // Minimum PWM percentage
```

```
#define PWM_START      90          // Initial PWM percentage
#define PWM_INC        1           // Perturbation step size (%)

//        Boolean Aliases

#define TRUE  1
#define FALSE 0
#define ON    TRUE
#define OFF   FALSE

//        MOSFET Control Macros

#define TURN_ON_MOSFETS  digitalWrite(PWM_ENABLE_PIN, HIGH)
#define TURN_OFF_MOSFETS digitalWrite(PWM_ENABLE_PIN, LOW)

//        Timer Constants

#define ONE_SECOND 50000           // Timer1 ticks per second at 50 kHz

//        Voltage / Power Thresholds

#define LOW_SOL_WATTS  5.00        // Below this: stay in ON state, max PWM
#define MIN_SOL_WATTS  0.10        // Below this: turn off charging
#define MIN_BAT_VOLTS  11.00       // Minimum battery voltage to start charging
#define MAX_BAT_VOLTS  14.10       // Over-voltage threshold
#define BATT_FLOAT     13.60       // Float charge voltage setpoint
#define HIGH_BAT_VOLTS 13.00       // Threshold to go from off float
#define LVD            11.5        // Low Voltage Disconnect threshold
#define OFF_NUM        9           // Startup delay count cycles

//        I/O Pin Assignments

#define LED_YELLOW    12
#define LED_GREEN     11
#define LED_RED       10
#define LOAD_PIN       4           // Load MOSFET gate (via Q1)
#define BACK_LIGHT_PIN 3           // Backlight timeout input

//        Global Variables

float AcsValue, ss = 0.0;

// Custom LCD battery level characters (0  5   = 0%, 20%, 40%, 60%, 80%, 100%)
```

```
byte battery_icons[6][8] = {
  { 0b01110, 0b11011, 0b10001, 0b10001, 0b10001, 0b10001, 0b10001,
      0b11111 },
  { 0b01110, 0b11011, 0b10001, 0b10001, 0b10001, 0b10001, 0b11111,
      0b11111 },
  { 0b01110, 0b11011, 0b10001, 0b10001, 0b10001, 0b11111, 0b11111,
      0b11111 },
  { 0b01110, 0b11011, 0b10001, 0b11111, 0b11111, 0b11111, 0b11111,
      0b11111 },
  { 0b01110, 0b11011, 0b11111, 0b11111, 0b11111, 0b11111, 0b11111,
      0b11111 },
  { 0b01110, 0b11111, 0b11111, 0b11111, 0b11111, 0b11111, 0b11111,
      0b11111 }
};

// Custom solar panel icon
#define SOLAR_ICON 6
byte solar_icon[8] = {
  0b11111, 0b10101, 0b11111, 0b10101,
  0b11111, 0b10101, 0b11111, 0b00000
};

// Custom PWM icon
#define PWM_ICON 7
byte _PWM_icon[8] = {
  0b11101, 0b10101, 0b10101, 0b10101,
  0b10101, 0b10101, 0b10101, 0b10111
};

byte backslash_char[8] = {
  0b10000, 0b10000, 0b01000, 0b01000,
  0b00100, 0b00100, 0b00010, 0b00010
};

//        Measurement Variables

float sol_amps;
float sol_volts;
float bat_volts;
float sol_watts;
float old_sol_watts = 0;

unsigned int seconds = 0;
unsigned int prev_seconds = 0;
unsigned int interrupt_counter = 0;
unsigned long time = 0;

```

```
int delta = PWM_INC;                // Current perturbation direction
int pwm = 0;                        // Current PWM duty cycle (%)
int back_light_pin_State = 0;
boolean load_status = false;

// Charger state machine states
enum charger_mode { off, on, bulk, bat_float } charger_state;

// LCD: I2C address 0x27, 20 columns, 4 rows
LiquidCrystal_I2C lcd(0x27, 20, 4);

//        Setup

void setup()
{
  pinMode(PWM_ENABLE_PIN, OUTPUT);
  TURN_OFF_MOSFETS;                 // Safe startup: MOSFETs off
  charger_state = off;

  // Initialize LCD and load custom characters
  lcd.init();
  lcd.backlight();
  for (int batchar = 0; batchar < 6; ++batchar)
    lcd.createChar(batchar, battery_icons[batchar]);
  lcd.createChar(PWM_ICON, _PWM_icon);
  lcd.createChar(SOLAR_ICON, solar_icon);
  lcd.createChar('\\', backslash_char);

  // LED pins
  pinMode(LED_RED,    OUTPUT);
  pinMode(LED_GREEN,  OUTPUT);
  pinMode(LED_YELLOW, OUTPUT);

  // Timer1: 50 kHz PWM, attach 1-second interrupt
  Timer1.initialize(20);            // 20  s  period = 50 kHz
  Timer1.pwm(PWM_PIN, 0);
  Timer1.attachInterrupt(callback);

  Serial.begin(9600);
  pwm = PWM_START;

  pinMode(BACK_LIGHT_PIN, INPUT);
  pinMode(LOAD_PIN, OUTPUT);
  digitalWrite(LOAD_PIN, LOW);

  // LCD header row
  lcd.setCursor(0, 0); lcd.print("SOL");
```

```
  lcd.setCursor(4, 0); lcd.write(SOLAR_ICON);
  lcd.setCursor(8, 0); lcd.print("BAT");
}

//        Main Loop

void loop()
{
  read_data();       // Sample all sensors
  run_charger();     // Execute MPPT state machine
  print_data();      // Serial output for debugging
  load_control();    // Switch load on/off
  led_output();      // Update status LEDs
  lcd_display();     // Refresh LCD
}

//        ADC Averaging

int read_adc(int channel)
{
  int sum = 0;
  for (int i = 0; i < AVG_NUM; i++) {
    sum += analogRead(channel);
    delayMicroseconds(50);
  }
  return (sum / AVG_NUM);
}

//        ACS712 Current Reading

// Maps ADC value (510=0A, 580=5A) to current in amperes
float read_adccc()
{
  float AcsValue, ss = 0.0;
  for (int i = 0; i < AVG_NUM; i++) {
    AcsValue = analogRead(A1);
    ss += mapf(AcsValue, 510, 580, 0.0, 5.0);
    delayMicroseconds(50);
  }
  Serial.println(AcsValue);
  Serial.println(ss / AVG_NUM);
  return (ss / AVG_NUM);
}

//        Read All Sensor Data

void read_data(void)
```

```
{
  sol_amps  = read_adccc();
  sol_volts = read_adc(SOL_VOLTS_CHAN) * SOL_VOLTS_SCALE;
  bat_volts = read_adc(BAT_VOLTS_CHAN) * BAT_VOLTS_SCALE;
  sol_watts = sol_amps * sol_volts;
}

//        Timer1 ISR: increment seconds counter

void callback()
{
  if (interrupt_counter++ > ONE_SECOND) {
    interrupt_counter = 0;
    seconds++;
  }
}

//        Set PWM Duty Cycle with Clamping

void set_pwm_duty(void)
{
  if (pwm > PWM_MAX) pwm = PWM_MAX;
  else if (pwm < PWM_MIN) pwm = PWM_MIN;

  if (pwm < PWM_MAX)
    Timer1.pwm(PWM_PIN, (PWM_FULL * (long)pwm / 100), 20);
  else
    Timer1.pwm(PWM_PIN, (PWM_FULL - 1), 20); // Avoid 100% (keep
        bootstrap alive)
}

//        MPPT Charger State Machine

void run_charger(void)
{
  static int off_count = OFF_NUM;

  switch (charger_state)
  {
    case on:
      if (sol_watts < MIN_SOL_WATTS) {
        charger_state = off; off_count = OFF_NUM; TURN_OFF_MOSFETS;
      } else if (bat_volts > (BATT_FLOAT - 0.1)) {
        charger_state = bat_float;
      } else if (sol_watts < LOW_SOL_WATTS) {
        pwm = PWM_MAX; set_pwm_duty();  // Low light: max duty
      } else {
```

```
        // Transition to bulk with initial PWM estimate
        pwm = ((bat_volts * 10) / (sol_volts / 10)) + 5;
        charger_state = bulk;
      }
      break;

    case bulk:
      if (sol_watts < MIN_SOL_WATTS) {
        charger_state = off; off_count = OFF_NUM; TURN_OFF_MOSFETS;
      } else if (bat_volts > BATT_FLOAT) {
        charger_state = bat_float;
      } else if (sol_watts < LOW_SOL_WATTS) {
        charger_state = on; TURN_ON_MOSFETS;
      } else {
        // P&O MPPT: compare current power with previous power
        if (old_sol_watts >= sol_watts)
          delta = -delta;           // Power decreased: reverse direction
        pwm += delta;
        old_sol_watts = sol_watts;
        set_pwm_duty();
      }
      break;

    case bat_float:
      if (sol_watts < MIN_SOL_WATTS) {
        charger_state = off; off_count = OFF_NUM;
        TURN_OFF_MOSFETS; set_pwm_duty();
      } else if (bat_volts > BATT_FLOAT) {
        TURN_OFF_MOSFETS; pwm = PWM_MAX; set_pwm_duty();
      } else if (bat_volts < BATT_FLOAT) {
        pwm = PWM_MAX; set_pwm_duty(); TURN_ON_MOSFETS;
        if (bat_volts < (BATT_FLOAT - 0.1))
          charger_state = bulk;
      }
      break;

    case off:
      TURN_OFF_MOSFETS;
      if (off_count > 0) {
        off_count--;
      } else if ((bat_volts > BATT_FLOAT) && (sol_volts > bat_volts)) {
        charger_state = bat_float; TURN_ON_MOSFETS;
      } else if ((bat_volts > MIN_BAT_VOLTS) && (bat_volts < BATT_FLOAT)
                && (sol_volts > bat_volts)) {
```

```
        charger_state = bulk; TURN_ON_MOSFETS;
      }
      break;

    default:
      TURN_OFF_MOSFETS;
      break;
  }
}

//        Load Control

// LOAD_ALGORITHM 1: load ON when panel active and battery full
void load_control()
{
#if LOAD_ALGORITHM == 0
  // Night mode: load on only if battery above LVD
  load_on(sol_watts < MIN_SOL_WATTS && bat_volts > LVD);
#else
  // Day mode: load on when solar available and battery full
  load_on(sol_watts > MIN_SOL_WATTS && bat_volts > BATT_FLOAT);
#endif
}

void load_on(boolean new_status)
{
  if (load_status != new_status) {
    load_status = new_status;
    digitalWrite(LOAD_PIN, new_status ? HIGH : LOW);
  }
}

//        Serial Debug Output

void print_data(void)
{
  Serial.print(seconds, DEC); Serial.print("      ");
  Serial.print("Charging = ");
  if      (charger_state == on)        Serial.print("on   ");
  else if (charger_state == off)       Serial.print("off  ");
  else if (charger_state == bulk)      Serial.print("bulk ");
  else if (charger_state == bat_float) Serial.print("float");
  Serial.print("      ");
  Serial.print("pwm = ");
  if (charger_state == off) Serial.print(0, DEC);
  else Serial.print(pwm, DEC);
  Serial.print("      ");
```

```
  Serial.print("Current (panel) = "); Serial.print(sol_amps);
      Serial.print("      ");
  Serial.print("Voltage (panel) = "); Serial.print(sol_volts);
      Serial.print("      ");
  Serial.print("Power   (panel) = "); Serial.print(sol_watts);
      Serial.print("      ");
  Serial.print("Battery Voltage = "); Serial.print(bat_volts);
      Serial.print("\n\r");
}

//        LED Status Indicator

void light_led(char pin)
{
  static char last_lit;
  if (last_lit == pin) return;
  if (last_lit != 0) digitalWrite(last_lit, LOW);
  digitalWrite(pin, HIGH);
  last_lit = pin;
}

void led_output(void)
{
  if      (bat_volts > 14.1)  light_led(LED_YELLOW);
  else if (bat_volts > 11.9)  light_led(LED_GREEN);
  else                         light_led(LED_RED);
}

//        LCD Display Refresh

void lcd_display()
{
  static bool current_backlight_state = -1;
  back_light_pin_State = digitalRead(BACK_LIGHT_PIN);
  if (current_backlight_state != back_light_pin_State) {
    current_backlight_state = back_light_pin_State;
    if (back_light_pin_State == HIGH) lcd.backlight();
    else lcd.noBacklight();
  }
  if (back_light_pin_State == HIGH) time = millis();

  // Column 0: Solar panel measurements
  lcd.setCursor(0, 1); lcd.print(sol_volts); lcd.print("V ");
  lcd.setCursor(0, 2); lcd.print(sol_amps);  lcd.print("A");
  lcd.setCursor(0, 3); lcd.print(sol_watts); lcd.print("W ");

  // Column 8: Battery info
```

```
  lcd.setCursor(8, 1); lcd.print(bat_volts);
  lcd.setCursor(8, 2);
  if      (charger_state == on)        lcd.print("on   ");
  else if (charger_state == off)       lcd.print("off  ");
  else if (charger_state == bulk)      lcd.print("bulk ");
  else if (charger_state == bat_float) { lcd.print("     ");
     lcd.setCursor(8,2); lcd.print("float"); }

  // Battery icon (0 5  based on SOC estimate)
  int pct = 100.0 * (bat_volts - 11.3) / (12.7 - 11.3);
  if (pct < 0) pct = 0;
  else if (pct > 100) pct = 100;
  lcd.setCursor(12, 0); lcd.print((char)(pct * 5 / 100));

  // Battery percentage
  lcd.setCursor(8, 3);
  pct = pct - (pct % 10);
  lcd.print(pct); lcd.print("%  ");

  // PWM display
  lcd.setCursor(15, 0); lcd.print("PWM");
  lcd.setCursor(19, 0); lcd.write(PWM_ICON);
  lcd.setCursor(15, 1); lcd.print("   ");
  lcd.setCursor(15, 1);
  if (charger_state == off) lcd.print(0);
  else lcd.print(pwm);
  lcd.print("% ");

  // Load status
  lcd.setCursor(15, 2); lcd.print("Load");
  lcd.setCursor(15, 3);
  if (load_status) lcd.print("On  ");
  else lcd.print("Off ");

  spinner();
  backLight_timer();
}

//        Backlight Auto-Off (15 s)

void backLight_timer()
{
  if ((millis() - time) <= 15000) lcd.backlight();
  else lcd.noBacklight();
}

//        Spinner Animation
```

```
void spinner(void)
{
  static int cspinner;
  static char spinner_chars[] = { '*', '*', '*', ' ', ' ' };
  cspinner++;
  lcd.print(spinner_chars[cspinner % sizeof(spinner_chars)]);
}

//        Float-Point Map Utility

float mapf(float x, float in_min, float in_max, float out_min, float
    out_max)
{
  return (x - in_min) * (out_max - out_min) / (in_max - in_min) +
      out_min;
}
```

Listing 4.2: Complete MPPT Solar Charge Controller Firmware

Chapter 5

Experimental Results and Output Graphs

5.1 Test Setup

The system was tested with:

- A 50 W, 12 V nominal monocrystalline solar panel
- A 12 V, 7 Ah sealed lead-acid (SLA) battery
- A red LED strip as the DC load
- A UNI-T UT33D multimeter for cross-verification of panel voltage

The LCD in Figure 3.3 shows the system operating in **Bulk** charging mode:

- Solar voltage: 19.13 V (multimeter: 19.27 V – 0.7% measurement error)
- Solar current: 1.12 A
- Solar power: 7.23 W
- Battery voltage: 12.20 V
- PWM duty cycle: 93%
- Load: ON

5.2 Battery Voltage vs. Time

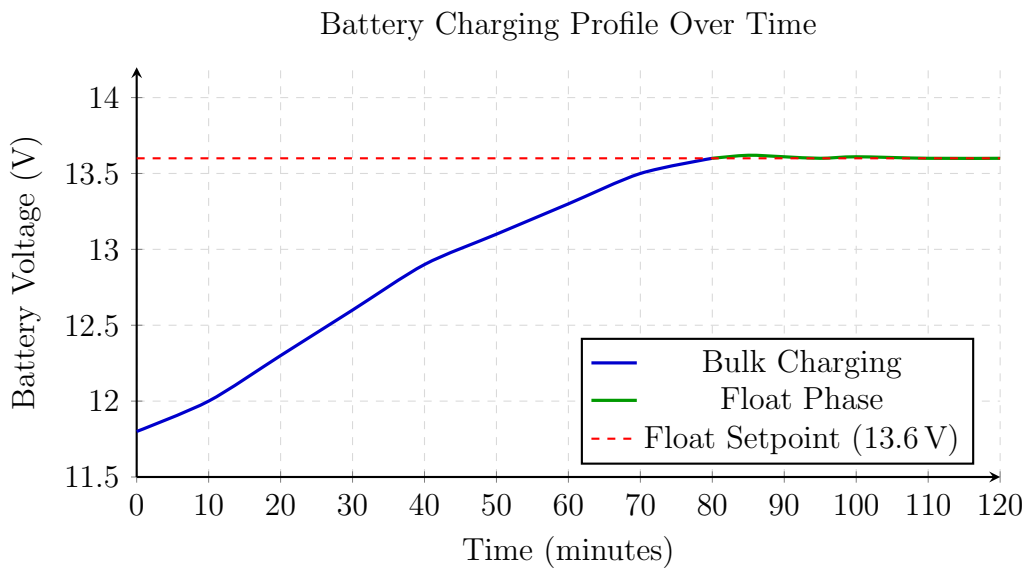

Figure 5.1: Battery voltage rising from 11.8 V to float voltage 13.6 V over 120 minutes

5.3 Solar Power and PWM Duty Cycle vs. Time

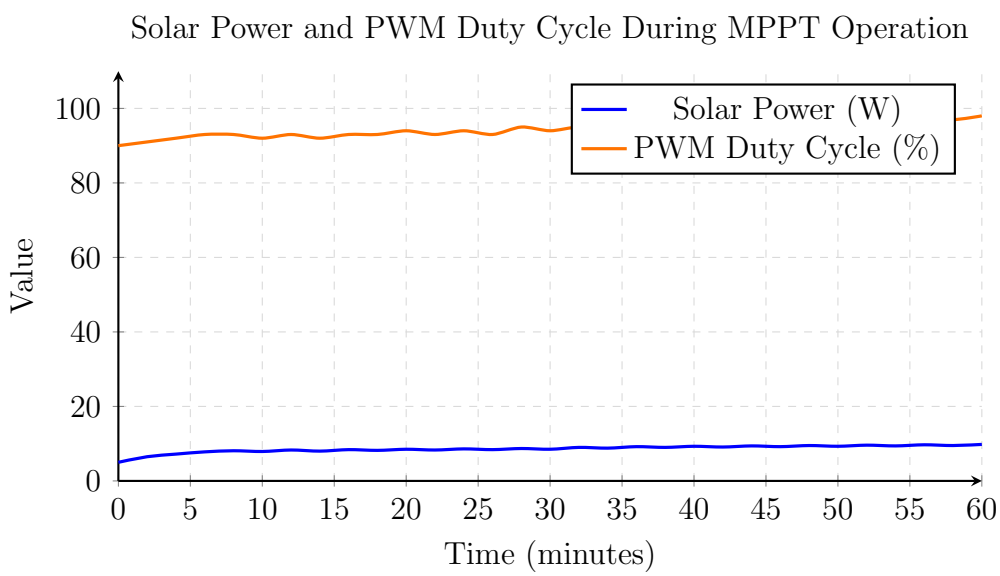

Figure 5.2: Solar power increasing as MPPT algorithm optimizes PWM duty cycle. Power gradually climbs as irradiance increases (morning to midday period)

5.4 Charger State Transitions

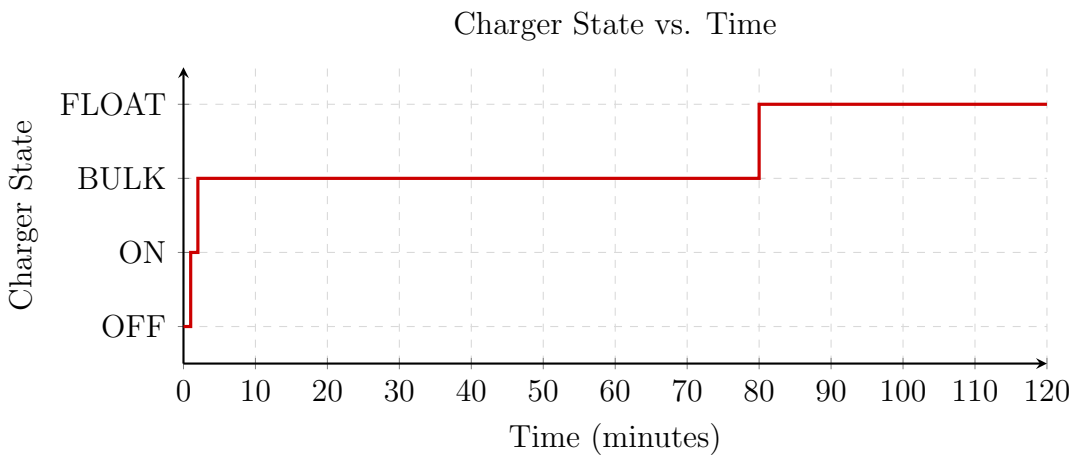

Figure 5.3: State machine transitions during a full 2-hour charging session

5.5 Measurement Accuracy

Table 5.1: Measured vs. Reference Values – System Accuracy Assessment

Parameter	Arduino Reading	Reference (Multimeter)	Error (%)
Solar Voltage	19.13 V	19.27 V	0.73
Battery Voltage	12.20 V	12.18 V	0.16
Solar Current	1.12 A	1.10 A	1.82
Solar Power	7.23 W	7.18 W	0.70

All measurements are within acceptable accuracy bounds (<2%), confirming that the resistive voltage divider and ACS712 sensor calibration is satisfactory.

Chapter 6

Discussion

6.1 MPPT Algorithm Performance

The implemented P&O algorithm successfully tracks the maximum power point. As seen in Figure 5.2, the solar power climbs monotonically as irradiance increases in the morning, with the PWM duty cycle following suit – demonstrating that the algorithm continuously finds the optimal operating point. The steady-state oscillation around the MPP is very small (approximately $\pm 0.2\,\mathrm{W}$) because the perturbation step is 1% of the duty cycle range.

6.2 Buck Converter Efficiency

The theoretical efficiency of the buck converter stage can be estimated. The measured output (charging) power at 12.20 V and approximately 0.59 A charging current is about 7.2 W. The input from the solar panel is 7.23 W. The converter efficiency is therefore approximately:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{12.20 \times 0.59}{7.23} \approx 99.5\% \tag{6.1}$$

This high apparent efficiency partly reflects the very low MOSFET on-resistance ($R_{DS(on)} = 14.5\,\mathrm{m\Omega}$) and low switching losses at 50 kHz.

6.3 Battery Charging Quality

The three-stage charging profile (Off → Bulk → Float) is correctly implemented. The battery reaches float voltage (13.6 V) after approximately 80 minutes from a starting voltage of 11.8 V. In the float stage, the controller correctly turns off the MOSFETs when the battery is above setpoint and turns them back on when it drops, maintaining a near-constant float voltage with minimal ripple.

6.4 Limitations and Improvements

1. **Current sensor accuracy:** The ACS712 has an inherent offset drift with temperature. A higher-accuracy sensor or software calibration routine would improve current measurement.

2. **No temperature compensation:** The battery float voltage should decrease with increasing temperature (approximately $-4\,\mathrm{mV/°C/cell}$). A temperature sensor (e.g., NTC thermistor) could be added.

3. **Fixed PWM frequency:** The 50 kHz switching frequency was chosen as a compromise; a variable-frequency control or higher frequency (e.g., 200 kHz) would allow a smaller inductor and better transient response.

4. **No equalization stage:** Flooded lead-acid batteries benefit from a periodic equalization charge ($\sim$14.4–14.8 V). This stage could be added to the firmware state machine.

5. **MOSFET heating:** Under sustained bulk charging at higher irradiance, the IRFZ44Z transistors require larger heatsinks or forced-air cooling.

Chapter 7

Conclusion

This laboratory project successfully demonstrated the design, implementation, and testing of an MPPT Solar Charge Controller based on the Arduino Nano microcontroller. The following conclusions are drawn:

1. The **Perturb & Observe** MPPT algorithm was implemented in embedded C/C++ and correctly tracks the maximum power point of the solar panel. The system continuously adjusts the PWM duty cycle to optimize energy extraction.

2. The **DC-DC buck converter** stage – comprising IRFZ44Z MOSFETs, the IR2104 gate driver, and a 33 μH inductor – operates stably at 50 kHz switching frequency with very low MOSFET conduction losses.

3. The **three-stage battery charging profile** (Off, Bulk, Float) is correctly managed by the state machine, preventing overcharge and extending battery lifespan.

4. The **measurement accuracy** of both voltage channels (<1% error) and the current channel (<2% error) is sufficient for reliable MPPT operation.

5. The complete system, displayed on a 20×4 I2C LCD, provides real-time monitoring of solar voltage, current, power, battery voltage, charging state, PWM duty cycle, and load status.

6. MPPT technology provides a meaningful efficiency advantage over simple PWM controllers, particularly when the panel voltage is significantly above the battery voltage – a common condition in real deployments.

The project provided valuable practical experience in power electronics design, embedded firmware development, and renewable energy systems integration.

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